



National University
of Computer & Emerging Sciences
Islamabad



MT- 1008 MULTIVARIABLE CALCULUS

Assignment No: 02

Individual Assignment

Semester: Spring 2024

Marks: 11x10

Due date: 24th March 2024, 11:59pm

Section: BSCS all sections

Instructors: Dr Hamda, Dr Imran Shahzad and Mr. M Arif

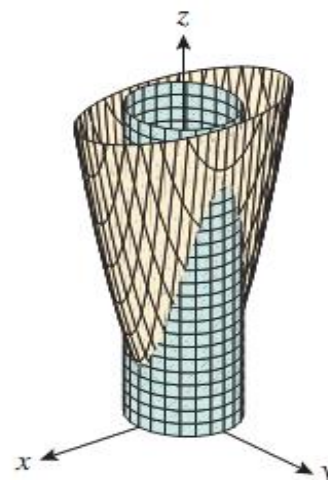
Instructions:

1. Plagiarized work will result in zero marks.
 2. No retake or late submission will be accepted.
 3. Attach complete code, results, and screenshot for questions that require programming solution.
 4. Questions that show the icon  require partial or complete solution using PYTHON.
 5. Submit the assignment on google classroom before the deadline.
 6. You must submit the assignment in a compiled form, i.e., a single PDF file of your complete assignment including programming and non-programming related questions.
 7. The PDF file should be according to the following format: Id_section_A1 e.g., i22-123456 A2. A2 in the end denotes Assignment 2.
 8. The images of by-hand solution should be properly scanned. You can use any mobile application such as Cam Scanner or Adobe Scan for scanning. Each of these applications allow you to export pdf or image files which you can use to combine with your programming solutions.
 9. Do not attach direct images from the camera application of your mobile phone, or screenshots.
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Question 1:

The accompanying figure shows the intersection of the elliptic paraboloid $z = x^2 + 4y^2$ and the right circular cylinder $x^2 + y^2 = 1$. Use **Lagrange multipliers** to find the highest and lowest points on the curve of intersection.



Solution :

$$\begin{aligned} z &= x^2 + 4y^2; \quad x^2 + y^2 = 1 \Rightarrow x^2 + y^2 - 1 = 0 \\ F(x, y, \lambda) &= x^2 + 4y^2 - \lambda(x^2 + y^2 - 1) \\ F_x = 0 &= 2x - 2\lambda x = 2x(1 - \lambda) \therefore x = 0 \text{ or } \lambda = 1 \\ F_y = 0 &= 8y - 2\lambda y = 2y(4 - \lambda) \therefore y = 0 \text{ or } \lambda = 4 \end{aligned}$$

$$\begin{aligned} 1) \quad x^2 + y^2 - 1 \Big|_{x=0} &\Rightarrow y = \pm 1 \quad \therefore (0, 1) \text{ and } (0, -1) \\ 2) \quad x^2 + y^2 - 1 \Big|_{y=0} &\Rightarrow x = \pm 1 \quad \therefore (1, 0) \text{ and } (-1, 0) \end{aligned}$$

$$\begin{aligned} \therefore \text{Critical points are } &(0, 1, 4), (0, -1, 4), (1, 0, 1), (-1, 0, 1) \\ \text{Maximum points: } &(0, 1, 4) \text{ and } (0, -1, 4) \\ \text{Minimum points: } &(1, 0, 1) \text{ and } (-1, 0, 1) \end{aligned}$$



Question 2:

A highway passes by the small town of Las Cienegas. From Las Cienegas, the highway is 5 miles to the north and 3 miles to the east. Assume that the highway is straight as it passes through this region. The town wants to build an access road (at a cost of \$250,000 per mile) to connect the highway. What is the shortest possible distance from Las Cienegas to the highway, and what would be the minimum cost of the road?

Solution:

Start writing here

Q_2

Let d = distance from Town to highway

$$d = \sqrt{(x-0)^2 + (y-0)^2}$$
$$d = \sqrt{x^2 + y^2}$$

let $f(x, y) = d^2 = x^2 + y^2$

By using Lagrange Multiplier

$$F(x, y) = f(x, y) - \lambda g(x, y)$$
$$F(x, y) = (x^2 + y^2) - \lambda(5x + 3y - 15)$$

1. $F_x(x, y) = 2x - 5\lambda$

2. $F_y(x, y) = 2y - 3\lambda$

Diagram:

Equation of Highway

$$\frac{y-5}{0-5} = \frac{x-0}{3-0}$$
$$3(y-5) = -5x$$
$$3y - 15 = -5x$$
$$5x + 3y - 15 = 0 \quad \text{--- (1)}$$



$$F_x(x, y) = 0 \quad \text{and} \quad F_y(x, y) = 0$$

$$2x - 5\lambda = 0$$

$$2y - 3\lambda = 0$$

$$\frac{2x}{5} = \lambda$$

$$\frac{2y}{3} = \lambda$$

$$\Rightarrow \frac{2x}{5} = \frac{2y}{3}$$

$$\Rightarrow \boxed{3x - 5y = 0} \quad \text{--- (2)}$$

Solving eq ① and ② $x = \frac{75}{34}$, $y = \frac{45}{34}$

Distance from Town to $p(x, y) = \sqrt{\left(\frac{75}{34} - 0\right)^2 + \left(\frac{45}{34} - 0\right)^2}$

$$= \frac{15}{\sqrt{34}} \text{ units}$$

Cost of the road $= 250,000 \times \frac{15}{\sqrt{34}} \quad \text{Ans}$

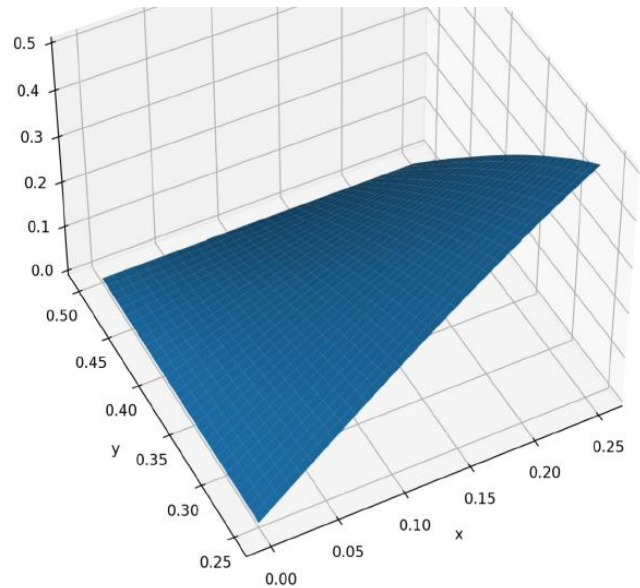


Question 3:

Let $f(x, y) = \sin \pi x \cos \pi y$ be a surface and R is the region described by $0 \leq x \leq \frac{1}{4}$, $\frac{1}{4} \leq y \leq \frac{1}{2}$.

(a) Draw the surface $f(x, y)$ by using Python.

```
1  from matplotlib import pyplot as plt
2  import numpy as np
3
4  x = np.linspace(0, 1.0/4, 32)
5  y = np.linspace(1.0/4, 1.0/2, 32)
6  x, y = np.meshgrid(x, y)
7  z = np.sin(np.pi * x) * np.cos(np.pi * y)
8
9  figure = plt.figure(figsize=(9, 4))
10
11 graph = plt.axes(projection = '3d')
12
13 graph.plot_surface(x, y, z)
14
15 plt.style.use('_mpl-gallery')
16
17
18 plt.title("Q3")
19 plt.xlabel("x")
20 plt.ylabel("y")
21 plt.show()
```



(b) Show that $0 \leq \iint_R f(x, y) dA \leq \frac{1}{32}$.

firstly we know

$$\sin 0 = 0 \quad \text{and} \quad \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\text{also} \quad \sin \theta \geq 0 \quad \text{if} \quad \theta \in [0, \frac{\pi}{4}] \quad \text{--- (1)}$$

$$\text{Similarly} \quad \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} \quad \text{and} \quad \cos \frac{\pi}{2} = 0$$

$$\cos \theta \geq 0 \quad \text{if} \quad \theta \in [\frac{\pi}{4}, \frac{\pi}{2}] \quad \text{--- (2)}$$

from (1) and (2), in given region R , $f(x, y)$ is positive



$$\text{Secondly} = \int_0^{1/4} \int_{1/4}^{1/2} \sin \pi x \cos \pi y \, dy \, dx$$

$$= \int_0^{1/4} \left[\sin \pi x \frac{\sin \pi y}{\pi} \right]_{1/4}^{1/2} dx$$

$$= \int_0^{1/4} \frac{\sin \pi x}{\pi} \left[\sin \frac{\pi}{2} - \sin \frac{\pi}{4} \right] dx$$

$$= \int_0^{1/4} \frac{\sin \pi x}{\pi} \left[1 - \frac{1}{\sqrt{2}} \right] dx$$

$$= \frac{\sqrt{2}-1}{\sqrt{2} \pi} \int_0^{1/4} \sin \pi x \, dx$$

$$= \frac{\sqrt{2}-1}{\sqrt{2} \pi} \left[-\frac{\cos \pi x}{\pi} \right]_0^{1/4}$$

$$= \frac{1-\sqrt{2}}{\sqrt{2} \pi^2} \left[\cos \pi x \right]_0^{1/4}$$

$$= \frac{1-\sqrt{2}}{\sqrt{2} \pi^2} \left[\cos \frac{\pi}{4} - \cos 0 \right]$$

$$= \frac{1-\sqrt{2}}{\sqrt{2} \pi^2} \left[\frac{1}{\sqrt{2}} - 1 \right]$$



Question 4:

Find the volume of the solid that is bounded by the elliptic paraboloid $x^2 + 2y^2 + z = 16$, the planes $x = 2$ and $y = 2$ and the three coordinate planes.

SOLUTION We first observe that S is the solid that lies under the surface $z = 16 - x^2 - 2y^2$ and above the square $R = [0, 2] \times [0, 2]$. (See Figure 5.) This solid was considered in Example 1 in Section 15.1, but we are now in a position to evaluate the double integral using Fubini's Theorem. Therefore

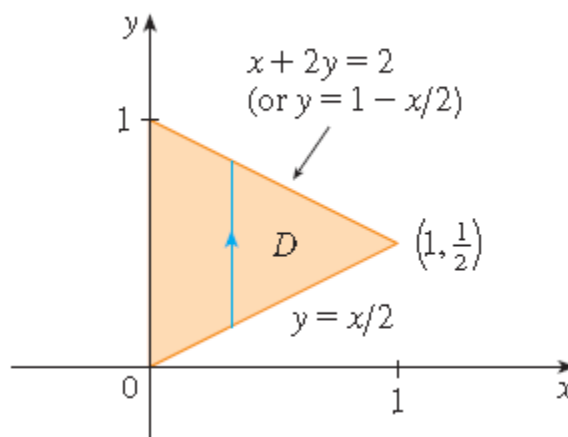
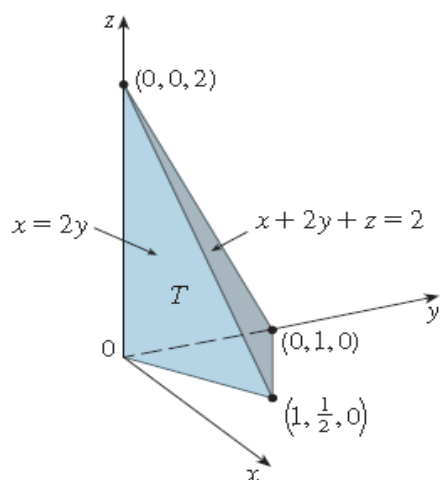
$$\begin{aligned} V &= \iiint_R (16 - x^2 - 2y^2) dA = \int_0^2 \int_0^2 (16 - x^2 - 2y^2) dx dy \\ &= \int_0^2 \left[16x - \frac{1}{3}x^3 - 2y^2x \right]_{x=0}^{x=2} dy \\ &= \int_0^2 \left(\frac{88}{3} - 4y^2 \right) dy = \left[\frac{88}{3}y - \frac{4}{3}y^3 \right]_0^2 = 48 \end{aligned}$$

□

Question 5:

Find the volume of the tetrahedron bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$, and $z = 0$.

Solution:





SOLUTION In a question such as this, it's wise to draw two diagrams: one of the three-dimensional solid and another of the plane region D over which it lies. Figure 13 shows the tetrahedron T bounded by the coordinate planes $x = 0$, $z = 0$, the vertical plane $x = 2y$, and the plane $x + 2y + z = 2$. Since the plane $x + 2y + z = 2$ intersects the xy -plane (whose equation is $z = 0$) in the line $x + 2y = 2$, we see that T lies above the triangular region D in the xy -plane bounded by the lines $x = 2y$, $x + 2y = 2$, and $x = 0$. (See Figure 14.)

The plane $x + 2y + z = 2$ can be written as $z = 2 - x - 2y$, so the required volume lies under the graph of the function $z = 2 - x - 2y$ and above

$$D = \{(x, y) \mid 0 \leq x \leq 1, x/2 \leq y \leq 1 - x/2\}$$

$$V = \iint_D (2 - x - 2y) \, dA = \int_0^1 \int_{x/2}^{1-x/2} (2 - x - 2y) \, dy \, dx$$

$$= \int_0^1 \left[2y - xy - y^2 \right]_{y=x/2}^{y=1-x/2} dx$$

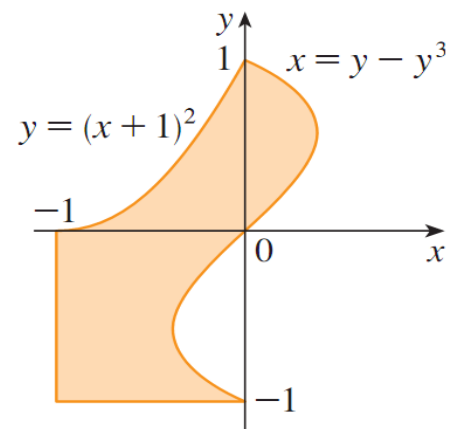
$$= \int_0^1 \left[2 - x - x \left(1 - \frac{x}{2} \right) - \left(1 - \frac{x}{2} \right)^2 - x + \frac{x^2}{2} + \frac{x^2}{4} \right] dx$$

$$= \int_0^1 (x^2 - 2x + 1) \, dx = \left[\frac{x^3}{3} - x^2 + x \right]_0^1 = \frac{1}{3}$$

□

Question 6:

By using double integral, find area of the shaded region given in the diagram.





Question 6:

$$A_1 = \text{Area} = \int_0^1 \int_{\sqrt{y}-1}^{y-y^3} dx dy \quad (\text{above } x\text{-axis})$$

$$A_2 = \text{Area} = \int_{-1}^0 \int_{-1}^{y-y^3} dx dy \quad (\text{below } x\text{-axis})$$

Solving A_1 :

$$\begin{aligned} &= \int_0^1 [x]_{\sqrt{y}-1}^{y-y^3} dy \\ &= \int_0^1 (y-y^3-\sqrt{y}+1) dy \\ &= \left[\frac{y^2}{2} - \frac{y^4}{4} - \frac{y^{3/2}}{3/2} + y \right]_0^1 \\ &= \frac{1}{2} - \frac{1}{4} - \frac{2}{3} + 1 = \frac{7}{12} \end{aligned}$$

Solving A_2 :

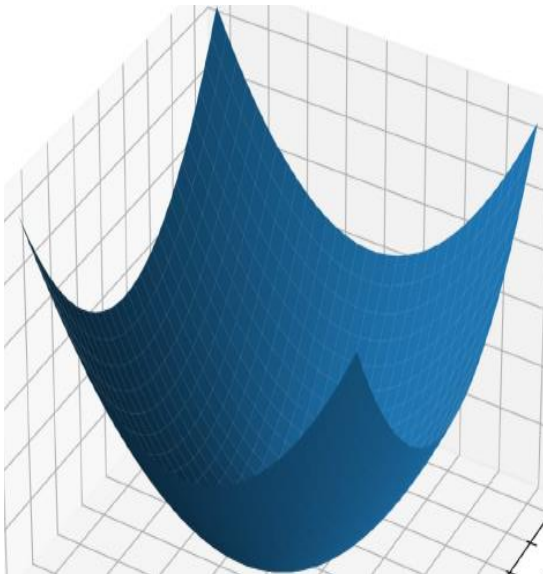
$$\begin{aligned} &= \int_{-1}^0 [x]_{-1}^{y-y^3} dy \\ &= \int_{-1}^0 (y-y^3+1) dy \\ &= \left[\frac{y^2}{2} - \frac{y^4}{4} + y \right]_{-1}^0 \\ &= (0-0+0) - \left(\frac{1}{2} - \frac{1}{4} - 1 \right) \\ &= -\frac{1}{2} + \frac{1}{4} + 1 \\ &= \frac{3}{4} \end{aligned}$$

Thus total area = $A_1 + A_2 = \frac{7}{12} + \frac{3}{4} = \frac{4}{3}$ sq unit Ans



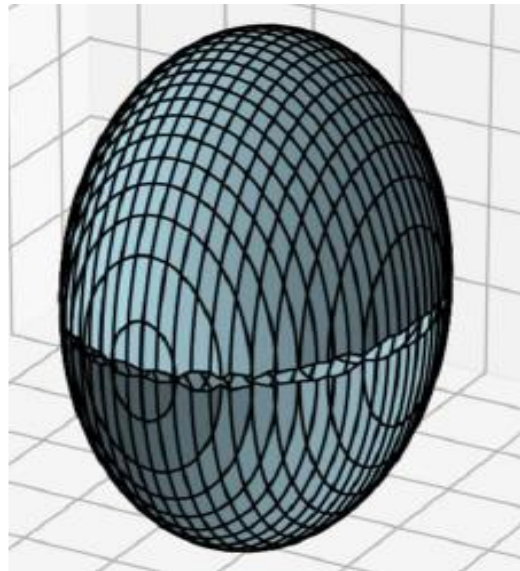
Question 7:

- (a) Using Python, draw the paraboloid $z = x^2 + y^2$ and the ellipsoid $3x^2 + 2y^2 + z^2 = 9$.



```
1 from matplotlib import pyplot as plt
2 import numpy as np
3
4 x = np.linspace(-4, 4, 32)
5 y = np.linspace(-4, 4, 32)
6 x, y = np.meshgrid(x, y)
7
8 z = x**2 + y**2
9
10 figure = plt.figure(figsize=(9, 4))
11 graph = plt.axes(projection = '3d')
12 graph.plot_surface(x, y, z)
13
14 plt.title("Q7: Paraboloid")
15 plt.xlabel("x")
16 plt.ylabel("y")
17 plt.show()
```

For Paraboloid



```
x = np.linspace(-4, 4, 5000)
y = np.linspace(-4, 4, 5000)
X, y = np.meshgrid(x, y)
z = f(x, y)

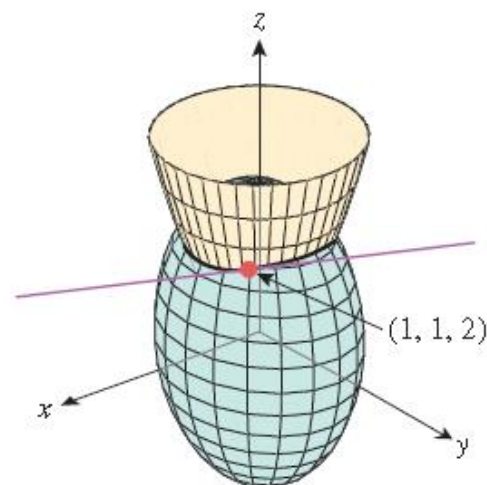
figure = plt.figure(figsize= (9, 4))
graph = plt.axes(projection = '3d')
graph.plot_surface(x, y, -z, color = 'lightblue', antialiased = True, edgecolor = 'black')
graph.plot_surface(x, y, z, color = 'lightblue', antialiased = True, edgecolor = 'black')

plt.title("Q7: Ellipsoid")
plt.xlabel("x")
plt.ylabel("y")
plt.show()
```

For ellipsoid



(b) Find parametric equations of the tangent line to the curve of intersection of the paraboloid $z = x^2 + y^2$ and the ellipsoid $3x^2 + 2y^2 + z^2 = 9$ at the point $(1, 1, 2)$.



Solution. We begin by rewriting the equations of the surfaces as

$$x^2 + y^2 - z = 0 \quad \text{and} \quad 3x^2 + 2y^2 + z^2 - 9 = 0$$

and we take

$$F(x, y, z) = x^2 + y^2 - z \quad \text{and} \quad G(x, y, z) = 3x^2 + 2y^2 + z^2 - 9$$

We will need the gradients of these functions at the point $(1, 1, 2)$. The computations are

$$\nabla F(x, y, z) = 2x\mathbf{i} + 2y\mathbf{j} - \mathbf{k}, \quad \nabla G(x, y, z) = 6x\mathbf{i} + 4y\mathbf{j} + 2z\mathbf{k}$$

$$\nabla F(1, 1, 2) = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}, \quad \nabla G(1, 1, 2) = 6\mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$$

Thus, a tangent vector at $(1, 1, 2)$ to the curve of intersection is

$$\nabla F(1, 1, 2) \times \nabla G(1, 1, 2) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 2 & -1 \\ 6 & 4 & 4 \end{vmatrix} = 12\mathbf{i} - 14\mathbf{j} - 4\mathbf{k}$$

Since any scalar multiple of this vector will do just as well, we can multiply by $\frac{1}{2}$ to reduce the size of the coefficients and use the vector of $6\mathbf{i} - 7\mathbf{j} - 2\mathbf{k}$ to determine the direction of the tangent line. This vector and the point $(1, 1, 2)$ yield the parametric equations

$$x = 1 + 6t, \quad y = 1 - 7t, \quad z = 2 - 2t \quad \blacktriangleleft$$

Since any scalar multiple of this vector will do just as well, we can multiply by $\frac{1}{2}$ to reduce the size of the coefficients and use the vector of $6\mathbf{i} - 7\mathbf{j} - 2\mathbf{k}$ to determine the direction of the tangent line. This vector and the point $(1, 1, 2)$ yield the parametric equations

$$x = 1 + 6t, \quad y = 1 - 7t, \quad z = 2 - 2t \quad \blacktriangleleft$$



Question 8:

- (a) Let $f(x, y) = xe^{xy}$ be a surface. Draw the surface $f(x, y)$ by using Python.

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

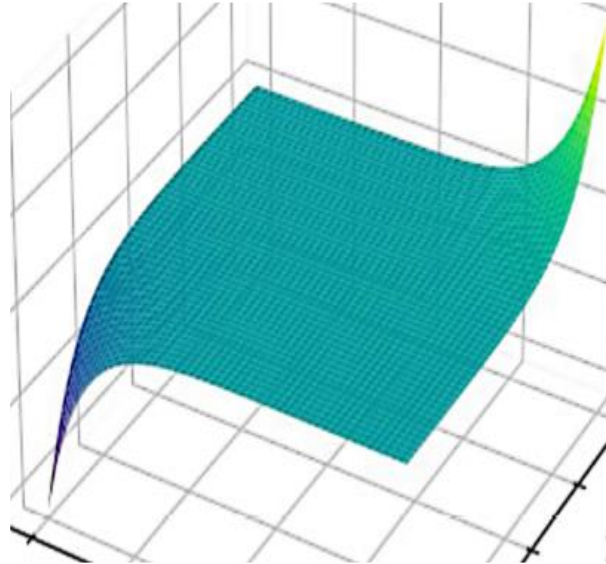
def f(x, y):
    return x * np.exp(x * y)

x = np.linspace(-2, 2, 100)
y = np.linspace(-2, 2, 100)
X, Y = np.meshgrid(x, y)
Z = f(X, Y)

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')
ax.plot_surface(X, Y, Z, cmap='viridis')

ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('f(x, y)')

plt.show()
```



- (b) Show that $f(x, y) = xe^{xy}$ is differentiable at $(1, 0)$ and find its **linearization** at $(1, 0)$. Then use it to approximate $f(1.1, -0.1)$.

SOLUTION The partial derivatives are

$$f_x(x, y) = e^{xy} + xye^{xy} \quad f_y(x, y) = x^2e^{xy}$$

$$f_x(1, 0) = 1 \quad f_y(1, 0) = 1$$

Both f_x and f_y are continuous functions, so f is differentiable by Theorem 8. The linearization is

$$\begin{aligned} L(x, y) &= f(1, 0) + f_x(1, 0)(x - 1) + f_y(1, 0)(y - 0) \\ &= 1 + 1(x - 1) + 1 \cdot y = x + y \end{aligned}$$

The corresponding linear approximation is

$$xe^{xy} \approx x + y$$

so
$$f(1.1, -0.1) \approx 1.1 - 0.1 = 1$$

Compare this with the actual value of $f(1.1, -0.1) = 1.1e^{-0.11} \approx 0.98542$.



Question 9:

- (a) If $z = f(x, y) = x^2 + 3xy - y^2$, find the differential dz .
(b) If x changes from 2 to 2.05 and y changes from 3 to 2.96, compare the values of Δz and dz .

SOLUTION

(a) Definition 10 gives

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = (2x + 3y) dx + (3x - 2y) dy$$

(b) Putting $x = 2$, $dx = \Delta x = 0.05$, $y = 3$, and $dy = \Delta y = -0.04$, we get

$$dz = [2(2) + 3(3)]0.05 + [3(2) - 2(3)](-0.04) = 0.65$$

The increment of z is

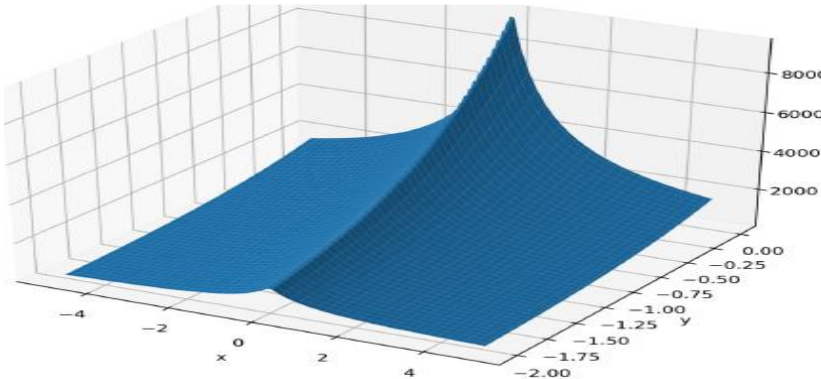
$$\begin{aligned}\Delta z &= f(2.05, 2.96) - f(2, 3) \\ &= [(2.05)^2 + 3(2.05)(2.96) - (2.96)^2] - [2^2 + 3(2)(3) - 3^2] \\ &= 0.6449\end{aligned}$$

Notice that $\Delta z \approx dz$ but dz is easier to compute.



Question 10:

If $f(x, y) = \frac{10,000e^y}{1 + \frac{|x|}{2}}$ represents the “population density” of a certain bacterium on the xy -plane, where x and y are measured in centimeters, find the total population of bacteria within the rectangle $-5 \leq x \leq 5$, $-2 \leq y \leq 0$.



$$\text{Total population} = \int_{-5}^5 \int_{-2}^0 \frac{10,000e^y}{1 + \frac{|x|}{2}} dy dx$$

$$= \int_{-5}^5 \left[\frac{10,000e^y}{1 + \frac{|x|}{2}} \right]_{-2}^0 dx$$

$$= \int_{-5}^5 \frac{10,000}{1 + \frac{|x|}{2}} [e^0 - e^{-2}] dx$$

$$= \frac{10,000(e^2 - 1)}{e^2} \int_{-5}^5 \frac{1}{1 + \frac{|x|}{2}} dx$$



$$= \frac{10,000(e^2-1)}{e^2} \left\{ \int_{-5}^0 \frac{1}{1+\frac{|x|}{2}} dx + \int_0^5 \frac{1}{1+\frac{|x|}{2}} dx \right\}$$

$$= \frac{10,000(e^2-1)}{e^2} \left\{ \int_{-5}^0 \frac{1}{1-\frac{x}{2}} dx + \int_0^5 \frac{1}{1+\frac{x}{2}} dx \right\}$$

$$= \frac{10,000(e^2-1)}{e^2} \left\{ \int_{-5}^0 \frac{2}{2-x} dx + \int_0^5 \frac{2}{2+x} dx \right\}$$

$$= \frac{10,000(e^2-1)}{e^2} \left\{ \left[-2 \ln|2-x| \right]_{-5}^0 + \left[2 \ln|2+x| \right]_0^5 \right\}$$

$$= \frac{10,000(e^2-1)}{e^2} \left\{ -2 \ln|2| + 2 \ln|7| + 2 \ln|7| - 2 \ln|2| \right\}$$

$$= \frac{10,000(e^2-1)}{e^2} \left\{ 4 \ln|7| - 4 \ln|2| \right\}$$

$$= \frac{40,000(e^2-1)}{e^2} \ln\left|\frac{7}{2}\right|$$



Question 11:

A swimming pool is circular with a 40 meter diameter. The depth is constant along east-west lines and increases linearly from 2 meters at the south end to 7 meters at the north end. Find the volume of the pool.

Sol

Since radius $r = 20$
Height on South $= 2$ ft
Height on North $= 7$ ft

South: $y = mx + c$
 $2 = m(-20) + c$
 $2 = -20m + c$ — (1)

North: $y = mx + c$
 $7 = m(20) + c$
 $7 = 20m + c$ — (2)

From (1) and (2) $c = 9/2$ $m = 1/8$

Thus height function $z = h(x, y) = \frac{1}{8}y + \frac{9}{2}$

$$\text{Volume of pool} = \iint_R h(x, y) dA$$
$$= \iint_R \left(\frac{1}{8}y + \frac{9}{2} \right) dA$$

Since pool is circular so we convert the integral into polar form

$$= \int_0^{2\pi} \int_0^{20} \left(\frac{1}{8}(r \sin \theta) + \frac{9}{2} \right) r dr d\theta$$

$$= \int_0^{2\pi} \int_0^{20} \left(\frac{1}{8}r^2 \sin \theta + \frac{9}{2}r \right) dr d\theta$$

$$= \int_0^{2\pi} \left(\frac{\sin \theta}{8} \frac{r^3}{3} + \frac{9}{2} \frac{r^2}{2} \right) \bigg|_0^{20} d\theta$$



$$= \int_0^{2\pi} \left(\frac{(20)^3 \sin \theta}{24} + \frac{400(9)}{4} \right) d\theta$$

$$= \frac{8000}{24} \left[-\cos \theta \right]_0^{2\pi} + 900 \left[\theta \right]_0^{2\pi}$$

$$= -\frac{8000}{24} \left[\overset{70}{\cancel{\cos 2\pi}} - \cos 0 \right] + 900 [2\pi - 0]$$

$$= 1800\pi \text{ cubic ft Ans}$$